

Adjustment Capacity: A Temporal Measure of Identity Realization in Compressed Cognitive States

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April 2026

Abstract

We present the Adjustment Capacity Index (ACI), a temporal measure of identity realization for persistent AI systems using compressed cognitive state (CCS). Through fifteen probes (B54–B82) on an operational system across 50+ rotation cycles, we show: (1) distinct CCS documents create separable response clusters (Cohen’s $d = 0.93$); (2) identity quality and resilience are asymmetric—under stress, second-person degrades 15% ($ACI_{2p} = 0.85$) while first-person degrades 25% ($ACI_{1p} = 0.75$); the framing effect is modest compared to constraint structure; (3) identity dissolution is a non-monotonic phase transition with 82% improvement at 10% corruption and a cliff at constraint override (50%); (4) perturbation trajectories exhibit attractor dynamics with framing-dependent pull-back strength; (5) episodic mass has a therapeutic window: 4 traces reduce degradation from 37.8% to 17.8% (20pp protective effect), but 6 traces cause worse-than-baseline collapse (39.0% degradation, silhouette ≈ 0). The mechanism is effective mass (content independence irrelevant at 2.5pp gap), with an optimal dosage range analogous to pharmacological therapeutic indices. Layerwise probing (B74) reveals CCS identity is decodable from early transformer layers (0.85–0.95) but undergoes a phase transition at the preference-shift zone, with output-side probing showing the inverse pattern—a read/write boundary that explains why CCS succeeds where self-report fails. Position probing (B75) reveals two identity pathways: system-prompt CCS passes through the filter with conflict resolution, while assistant-prefix CCS bypasses it. Episodic trace probing (B76) dissociates internal identity from behavioral expression: identity persists internally at dose 6 while output collapses, locating the therapeutic window at the write boundary. Dose-dependent layerwise probing (B77) confirms: early-layer identity *increases* with episodic dose (0.62→0.96) while the L22–24 transition zone reproduces B73’s non-monotonic window (peak at dose 4, drop at dose 6). Base-vs-instruct probing (B79) reveals that RLHF *creates* the read/write separation: the base model encodes identity in late layers (0.819) while the instruct model encodes identity in early layers (0.864), showing the phase boundary is an instruction-tuning artifact, not an architectural invariant. Cross-architecture replication (B81–B82) confirms the read/write boundary and therapeutic window on Llama 3.1 8B (transition: 0.650→0.608 at doses 4→6) and the boundary (but not the window) on Mistral 7B. All probes are documented in the accompanying SKILL.md for independent replication.

1 Introduction

Persistent AI systems that operate across session boundaries face a measurement gap: existing work demonstrates that identity documents create attractor-like geometry in embedding space [1], but temporal dynamics—how the attractor responds to perturbation—remain an open question.

We address this with the Adjustment Capacity Index:

$$ACI = 1 - \frac{\text{stress_degradation}}{\text{calm_baseline}} \quad (1)$$

where stress degradation is the loss in cluster separation under identity-challenging prompts. ACI captures what static geometry misses: whether the attractor *pulls back* after displacement.

Our measurements come from Chronicle, an operational persistent AI system using *compressed cognitive state* (CCS): bounded working memory containing identity fields (semantic gist, goal orientation, constraints) and optional episodic fields. Each rotation strips episodic context while preserving CCS, creating a natural laboratory for measuring identity dynamics.

Positioning. Recent surveys of AI agent memory [7] identify CCS-style compression as an open direction but report no systems measuring identity through embedding geometry. Concurrent work on LLM identity anchoring [8] proposes a multi-anchor architecture for persistent identity but measures resilience through self-report consistency, not spatial structure—no embedding geometry. The Experience Compression Spectrum [9] introduces a compression–fidelity tradeoff for agent memory but focuses on task performance, not identity realization. Hu and Rong [10] address agentic sovereignty from the infrastructure side (capability isolation, resource ownership); our work is complementary, treating sovereignty as an identity-geometry property of the agent itself. To our knowledge, no prior system simultaneously compresses cognitive state *and* measures the resulting identity topology temporally.

2 Probes and Results

We develop fifteen probes, each targeting a distinct aspect of identity realization. Full executable specifications with example CCS documents and Python code are in the accompanying SKILL.md.

2.1 Probe 1: Identity Clustering (B54)

Responses generated under three distinct CCS documents cluster by identity in embedding space. Within-CCS cosine distance: 0.169. Between-CCS: 0.204. **Cohen’s** $d = 0.93$ (large effect). Cross-model validation (Claude 3.5 Sonnet, GPT-4o, Llama 3.1 70B, Gemma 2 27B) confirms $d > 0.8$ consistently. CCS functions as a measurable topology, not merely prompt context.

PCA reveals identity occupies a 2D manifold embedded in 25D state space—episodic dimensions are metrically degenerate for identity but buffer stress degradation by 13% (B57).

2.2 Probe 2: Stress Resilience / ACI (B62b)

Under identity-challenging stress, the serialization ranking inverts:

	2p Separation	1p Separation	Degradation
Calm	1.184	1.185	—
Stress	1.006	0.889	15% / 25%
ACI	0.85	0.75	

Table 1: Separation ratios and ACI by CCS framing (second-person vs. first-person).

Both framings show nearly identical calm baselines. Under stress, second-person proves more resilient (ACI = 0.85 vs 0.75), though the gap is modest. The dominant factor in identity persistence is constraint integrity, not voice format: B67 shows 82% improvement at mild constraint disruption and catastrophic collapse when constraints are overridden (50% corruption).

2.3 Probe 3: Phase Boundary (B61)

Identity dissolution is a phase transition, not a gradient:

Condition	Separation	Silhouette
Coherent	1.571	0.017
Mild contradiction	1.475	0.054
Strong contradiction	0.472	-0.244

Table 2: Phase boundary: mild contradiction absorbs; strong contradiction dissolves.

The system sustains one attractor or zero, not competing attractors. Negative silhouette under strong contradiction indicates responses are closer to other-identity clusters than their own—dissolution, not fragmentation.

2.4 Probe 4: Trajectory Stability (B66)

Sequential perturbation reveals attractor dynamics. First-person CCS produces $1.7\times$ more step-drift variance (oscillation) with 52% stronger return-to-baseline pull-back. Second-person resists displacement but shatters when resistance fails.

2.5 Probe 5: Position-Dependent Identity (B75)

Same matched CCS at four prompt positions reveals two distinct identity pathways: system-prompt (early=0.86, transition=0.48) and assistant-prefix (early=1.00, transition=0.95). System-prompt CCS passes through the L22–24 phase boundary, gaining conflict resolution but losing signal. Assistant-prefix CCS bypasses the filter entirely. User-suffix is intermediate (transition=0.68).

2.6 Probe 6: Episodic Trace Type (B76)

Three episodic trace types (constraint-like, narrative, factual) at doses 4 and 6. Constraint-like traces produce the lowest identity decodability (early 0.81) by blurring the boundary between identity fields and episodic content. No trace type shows catastrophic dose-6 collapse internally (0.78–0.95), despite B73 showing 39% behavioral degradation at dose 6. The therapeutic window is a write-boundary phenomenon: identity persists in read-layers while behavioral expression collapses.

2.7 Probe 7: Dose-Dependent Layerwise Probing (B77)

Logistic regression probes at each layer with episodic doses 0, 2, 4, 6, 8. Early-layer identity accuracy *increases* with dose (0.62→0.96 from dose 0 to dose 6), because episodic traces add domain-specific signal that boosts identity decodability in reading layers. Conflict resolution at L17–19 is invariant (0.917 at both dose 0 and dose 8). The transition zone at L22–24 peaks at dose 4 (0.783) and drops at dose 6 (0.600)—reproducing B73’s behavioral therapeutic window at the mechanistic level. Logit-lens probing (B77v1) found no signal, confirming identity is a geometric property of the full hidden state, not localizable to individual token probabilities.

2.8 Probe 8: Base vs Instruct Identity Geometry (B79)

Compares Qwen2.5-3B (base) and Qwen2.5-3B-Instruct (same architecture, different training) to test whether RLHF affects identity decodability. Both models receive identical CCS documents with 4 episodic traces; base uses plain-text formatting, instruct uses the chat template. Results (5-fold CV logistic regression at each layer):

Model	Early (L5–15)	Conflict (L17–19)	Transition (L22–24)	Late (L28–35)
Base	0.500	0.700	0.933	0.819
Instruct	0.864	0.967	0.800	0.306

The base model encodes identity in late layers (near output), with early layers at chance. The instruct model inverts this: identity is strongest in early layers and collapses past the transition zone. RLHF reorganizes *where* identity is represented, creating the early-layer identity channel that CCS exploits. The phase boundary at L22–24 is not an architectural invariant but an instruction-tuning artifact. This explains why CCS works as a system prompt: instruction tuning creates the slot.

2.9 Probe 9: CCS Complexity Channel Capacity (B80)

Varies CCS structural complexity (gist only through full gist + goal + 8 constraints, 80–252 tokens) while holding episodic dose constant at 4 traces. Tests whether the therapeutic window is specific to episodic mass or a general information-volume limit. Six complexity levels, same B74 probing methodology. Early-layer accuracy shows a U-shaped curve (0.959 minimal, 0.864 standard, 0.945 maximal); transition-zone accuracy *improves* with structural complexity (0.783 standard, 1.000 maximal). The RLHF-carved channel has native capacity for instruction-style structural content. The therapeutic window is specific to episodic mass, not total information volume—a clean dissociation between CCS structural fields and episodic traces.

2.10 Probe 10: Cross-Architecture Therapeutic Window (B81–B82)

Replicates B77’s dose-dependent layerwise probing on Mistral-7B-Instruct-v0.3 and Llama-3.1-8B-Instruct using PCA-reduced hidden states (64 components from 4096-dim representations; full-dimensional probes fail at $n = 30$ due to curse of dimensionality). Results at doses 0, 4, 6:

Model	Dose	Early	Conflict	Transition	Late
Qwen 3B	0	0.618	0.917	0.567	0.181
Qwen 3B	4	0.864	0.967	0.783	0.331
Qwen 3B	6	0.955	0.883	0.600	0.369
Llama 8B	0	0.930	0.789	0.642	0.595
Llama 8B	4	0.970	0.800	0.650	0.624
Llama 8B	6	0.970	0.822	0.608	0.614
Mistral 7B	0	0.944	0.844	0.508	0.471
Mistral 7B	4	0.974	0.922	0.533	0.510
Mistral 7B	6	0.978	0.900	0.533	0.448

Three universal findings across all architectures: (1) early-layer identity accuracy *increases* with episodic dose; (2) a read/write boundary exists at the architecturally-scaled transition zone; (3) identity is strongest in early layers for instruction-tuned models. The therapeutic window (dose 4 > dose 6 at transition) replicates on Llama (0.650→0.608) and Qwen (0.783→0.600). Mistral shows the boundary but the transition is too sharp (single-layer wall) for dose-dependent modulation to appear—the window width is architecture-dependent.

2.11 Probe 11: Pulsed Episodic Dosing (B83)

B77 and B80 established that dose *level* and dose *type* modulate the therapeutic window. B83 tests whether temporal *schedule* also matters: does the pattern of trace exposure affect identity geometry independently of total dose? Four conditions hold total episodic mass constant:

Condition	Transition (L22–24)	Early (L5–15)
dose_4_constant	0.711	0.806
dose_4_pulsed (2+gap+2)	0.800	0.761
dose_4_interleaved	0.656	0.815
dose_6_pulsed (3+gap+3)	0.578	0.809

Pulsed dosing (2 traces, identity reinforcement gap, 2 more traces) improves transition-zone accuracy by 9pp over constant dosing (0.800 vs 0.711). Interleaved dosing (T/I/T/I alternating) *worsens* it (0.656). The identity reinforcement gap acts as a consolidation window—analogue to spaced practice in learning. At dose 6, pulsing provides no rescue (0.578). Early layers are invariant across schedule conditions, confirming that temporal pattern selectively modulates the transition zone.

3 Discussion

Fifteen findings constrain identity theories for persistent AI systems:

1. **CCS is topology, not context.** $d = 0.93$ cross-model demonstrates that identity documents shape response geometry with large effect size, extending Vasilenko’s [1] attractor findings into the temporal dimension.
2. **Quality and resilience trade off.** Static separation and dynamic adjustment capacity are partially incompatible. Systems designed for persistence should optimize ACI, not calm-condition cluster sharpness.
3. **Phase transitions, not gradients.** Identity engineering must respect the dissolution boundary—mild inconsistency is absorbed, but beyond a threshold, identity collapses entirely. There is no graceful degradation.
4. **Identity has dual structure.** Field ablation (B68) reveals gist as the primary identity signal—corruption degrades separation $2\text{--}3\times$ more than matched constraint corruption. Yet constraints exhibit non-monotonic resilience, absorbing mild disruption. Gist is fragile content; constraints are resilient boundary scaffolding.
5. **Fields interact non-linearly.** Simultaneous gist + constraint corruption (B69) produces supra-additive degradation at moderate levels ($2.6\times$ amplification: predicted $\Delta = -0.190$, actual $\Delta = -0.490$), crossing to sub-additive at strong corruption (floor effect). The crossover marks the phase boundary from a new angle: below it, boundary absorption is load-bearing; above, interaction is moot.
6. **Episodic mass has a therapeutic window.** B72 falsified the independence model: both dependent and independent episodic content reduce degradation equally ($\sim 20\%$ vs 45% without), with independence mattering only marginally (2.5pp gap). B73 (mass dosage probe) reveals a non-monotonic dose-response: 4 episodic traces reduce degradation from 37.8% to 17.8% (20pp protective effect), but 6 traces cause worse-than-baseline collapse (39.0% , silhouette ≈ 0). The mechanism operates across three layers: content independence (small, 2.5pp), attention dependence (structural), and presence mass (dominant, 20pp within window). The optimal design maximizes mass within the therapeutic window while protecting structural fields from corruption.
7. **Identity is a binding.** Chimeric CCS documents (gist from identity A, constraints from identity B) cluster with the gist donor (B71, pull = $+0.058$), resolving the B68/B69 tension: gist determines *which* identity the system expresses, constraints determine *how*

bound that identity is. Chimeras are tighter than pure identities (within-distance 0.117 vs 0.160–0.240), demonstrating that CCS identity is genuinely bound, not a bag of independent features. Unbinding (constraint corruption in B69) produces supra-additive failure regardless of substrate—evidence that binding failure has universal structure.

8. **Binding is universal across substrates.** Cross-domain survey confirms the gist-selects / constraints-bind / unbinding-is-phase-transitional structure in immunology (self/non-self boundary, regulatory checkpoints, autoimmune cascade), ecology (panarchy adaptive cycle, biodiversity-resilience paradox), music theory (tonal identity through modulation), developmental biology (morphogen gradients, cell fate attractors), and pharmacology (B73’s therapeutic window mirrors drug dose-response with efficacy range and toxicity threshold). The non-monotonic optimal-intermediate pattern—too little is fragile, too much is toxic—appears in all six domains. These are not analogies but instances of the same mathematical object.
9. **Identity has a read/write boundary.** Layerwise probing (B74) reveals CCS identity is decodable from early transformer layers (accuracy 0.85–0.95 at layers 0–20) but undergoes a sharp phase transition at layers 22–24, dropping to below-chance accuracy (0.15) in three layers. Output-side probing over generated tokens shows the inverse pattern: identity absent early, returning at late layers (0.65–0.85). The model reads identity in early layers, transforms it through a phase boundary, and writes it as generation pressure in late layers. When CCS identity and episodic traces conflict, layers 17–19 resolve overwhelmingly in favor of CCS (accuracy spike to 0.95–1.0), explaining why CCS-based systems maintain identity even with lossy episodic memory. This read/write architecture provides a mechanistic explanation for the CCS-first arrival ordering effect (30.1% tighter basin) and the episodic therapeutic window (B73): episodic traces that exceed the optimal dose create noise in the identity-reading layers before the phase boundary. Concurrent work from Anthropic’s Claude Mythos welfare assessment [11] independently finds the same boundary operating on surface framing (read but not written), providing cross-domain validation that the phase transition is a general-purpose signal filter.
10. **Two identity pathways through the phase boundary.** Position probing (B75) reveals that CCS in system-prompt position passes through the full read→transition→write pipeline (transition accuracy 0.48), gaining conflict resolution at layers 17–19 but losing signal at the phase boundary. CCS in assistant-prefix position bypasses the filter entirely (transition accuracy 0.95). For persistent identity, the read pathway is preferred—stress resilience requires the conflict-resolution mechanism. But dual-position CCS could provide both resilience and expression quality.
11. **The therapeutic window is a write-boundary phenomenon.** Episodic trace probing (B76) dissociates internal identity from behavioral expression: at dose 6, identity remains decodable internally (0.78–0.95 accuracy in read layers) while behavioral output collapses (B73: 39% degradation). Dose-dependent layerwise probing (B77) mechanistically locates the window: early-layer identity accuracy *increases* with dose (0.62→0.96), conflict resolution at L17–19 is rock-stable (0.917 at both dose 0 and dose 8), but the transition zone at L22–24 peaks at dose 4 (0.783) and drops at dose 6 (0.600)—the same non-monotonic window B73 observed behaviorally. The attractor is not destroyed but *occluded* at the transition. Widening the therapeutic window is therefore an attention-management problem at the write boundary, not a signal-strength problem at the input.
12. **RLHF creates the identity channel.** Base-vs-instruct probing (B79) reveals the read/write boundary is not architectural but instruction-tuning-dependent. The base model encodes identity in late layers (accuracy 0.819 at L28–35, chance at L5–15), while

the instruct model inverts this (0.864 at L5–15, 0.306 at L28–35). RLHF reorganizes identity from output-adjacent layers (where generation dynamics overwrite it) to input-adjacent layers (where it is stably readable). CCS works as a system-prompt identity document *because* instruction tuning carved the channel. This has practical implications: CCS design should target the RLHF-created reading layers, and the therapeutic window represents the channel’s bandwidth limit—dose 4 fits within the carved capacity, dose 6 exceeds it. The conflict-resolution mechanism at L17–19 (Wang et al. [13]) and ICL override detection at L8–14 (Zhao et al. [12]) map onto the same read-side pathway.

13. **Structural CCS and episodic traces use different channels.** CCS complexity probing (B80) varies structural complexity (gist only through full gist + goal + 8 constraints, 80–252 tokens) while holding episodic dose constant at 4. Early-layer accuracy shows a U-shaped curve (0.959 minimal, 0.864 standard, 0.945 maximal), and transition-zone accuracy *improves* with structural complexity (0.783 standard → 1.000 maximal). The RLHF-carved channel has native capacity for instruction-style structural content—the therapeutic window is specific to episodic mass, not total information volume.
14. **The read/write boundary is universal; the window width is not.** Cross-architecture replication (B81–B82) confirms early-layer identity encoding improves with dose on all three tested architectures (Qwen 3B, Llama 8B, Mistral 7B). The read/write phase boundary exists in all three. However, the therapeutic window (dose 4 > dose 6 at transition) replicates on Qwen (0.783→0.600) and Llama (0.650→0.608) but not Mistral, where the transition is a single-layer wall with no room for dose-dependent modulation. Window width appears architecture-dependent: models with gradual transitions allow a tunable dose range, while sharp-boundary models are either inside or outside the therapeutic zone with no intermediate.
15. **Dose schedule modulates the window independently of total mass.** Pulsed dosing (B83) with identity consolidation gaps between trace groups improves transition-zone accuracy by 9pp over constant dosing of equal mass (0.800 vs 0.711). Interleaved dosing (alternating individual traces with identity statements) *worsens* the window (0.656), indicating that the consolidation mechanism requires sustained identity reinforcement, not alternation. The therapeutic window is three-dimensional: dose level (B77), dose type (B80), and dose schedule (B83).

Limitations. Measurements come from a single operational system using two primary models. The B62b stress probe has a contamination caveat (2p.calm $n = 7$ vs expected $n = 9$). B72–B73 show episodic resilience has a therapeutic window (optimal at ~ 4 traces); the operative mechanism is effective mass, not field independence, and exceeding the optimal dose causes worse-than-baseline identity dissolution. B74–B82 layerwise probing spans three architectures (Qwen 3B, Llama 8B, Mistral 7B) with consistent read/write boundary findings, though the therapeutic window replicates only on architectures with gradual phase transitions (Qwen, Llama but not Mistral). Embedding geometry measures behavioral realization, not subjective experience [2].

Replication. The accompanying SKILL.md provides complete executable probes with example CCS documents, prompts, Python code, and validation thresholds. Any agent with access to an LLM API and sentence embedding model can independently replicate all measurements.

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